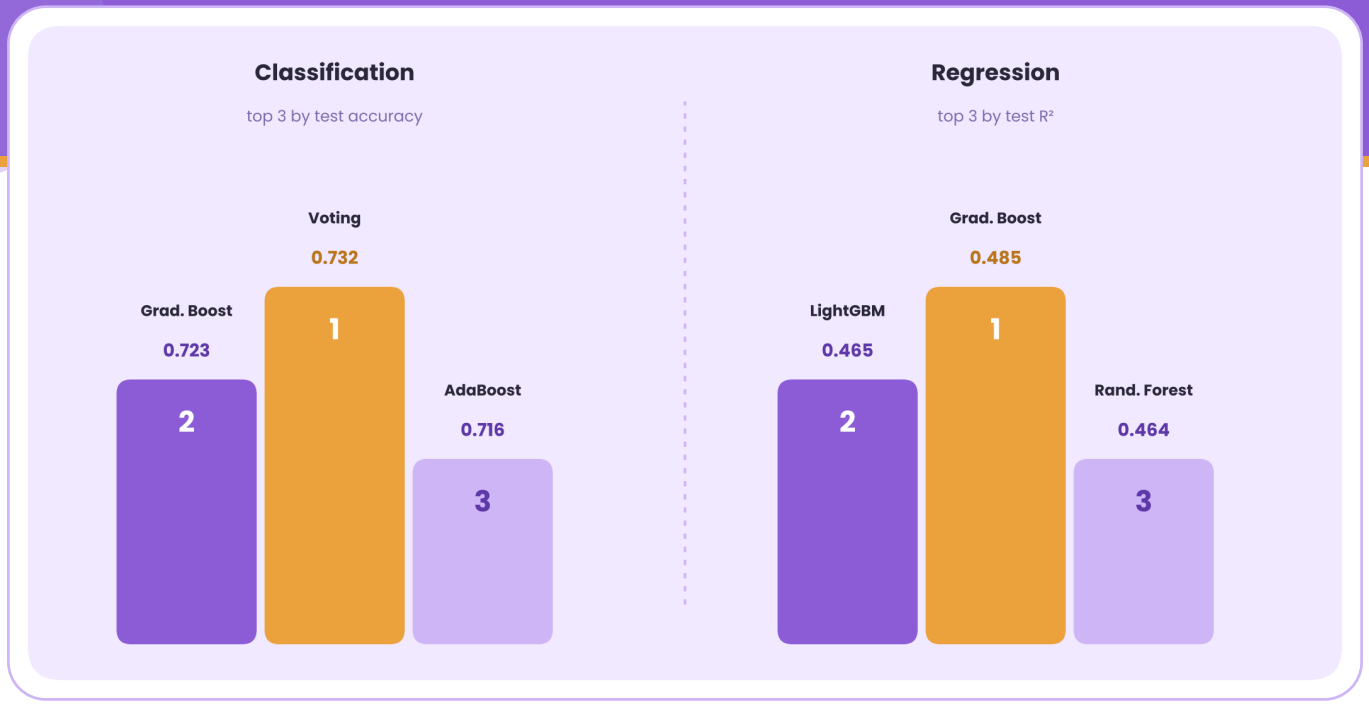
SUPERVISED MACHINE LEARNING

# SMART OUTCOME PREDICTOR

**Part G: Final Analysis & Reporting**

## Compare. Recommend. Deploy.

A clear read of what the notebook results show about bagging, boosting, voting and stacking, and which models to put into deployment.




**5,200**
students in the dataset

**15**
input features

**80 / 20**
train / test split (4,160 | 1,040)

**01 Compare**
Bagging vs boosting

**02 Evaluate**
Boosting algorithms

**03 Deploy**
Final recommendation

## Q Prepare a final report covering the points below.

*Every score here is a test-set result on 1,040 students. Classification is judged by accuracy and regression by $R^2$ score.*

## 1 Impact of bagging vs boosting

Bagging fits many models in parallel on bootstrap samples and averages them, which chiefly cuts variance. Boosting fits models in sequence so each one repairs the last one's mistakes, which chiefly cuts bias.

| | |
|---|---|
| **Against a single model** | One decision tree reached 0.645 accuracy, and bagging lifted that to 0.698, roughly 5 points higher. |
| **Classification** | AdaBoost (0.716), Gradient Boosting (0.723) and LightGBM (0.712) all finished ahead of bagging (0.698), while XGBoost (0.681) fell short. |
| **Regression** | Bagging ($R^2$ 0.456) outscored AdaBoost (0.407) and XGBoost (0.390), but was itself beaten by Gradient Boosting (0.485) and LightGBM (0.465). |
| **Overall** | Gradient Boosting was the strongest bagging-or-boosting model on both tasks, yet boosting did not win automatically: some algorithms trailed bagging. |

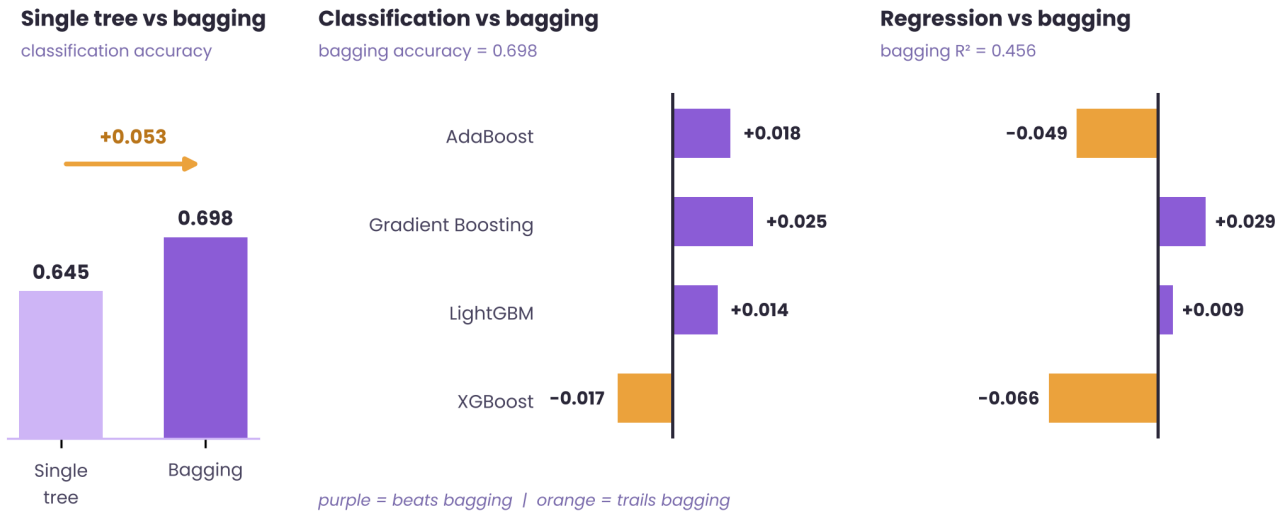
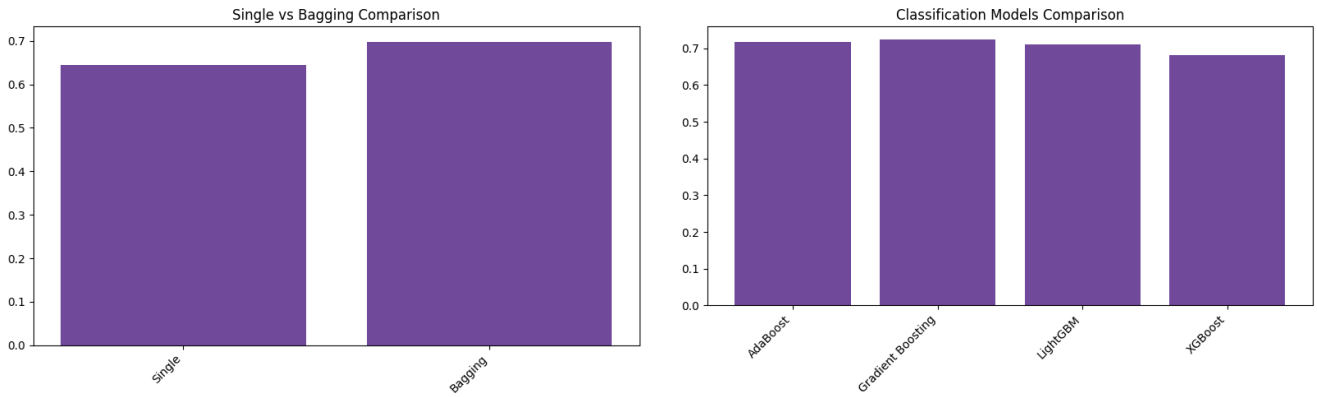
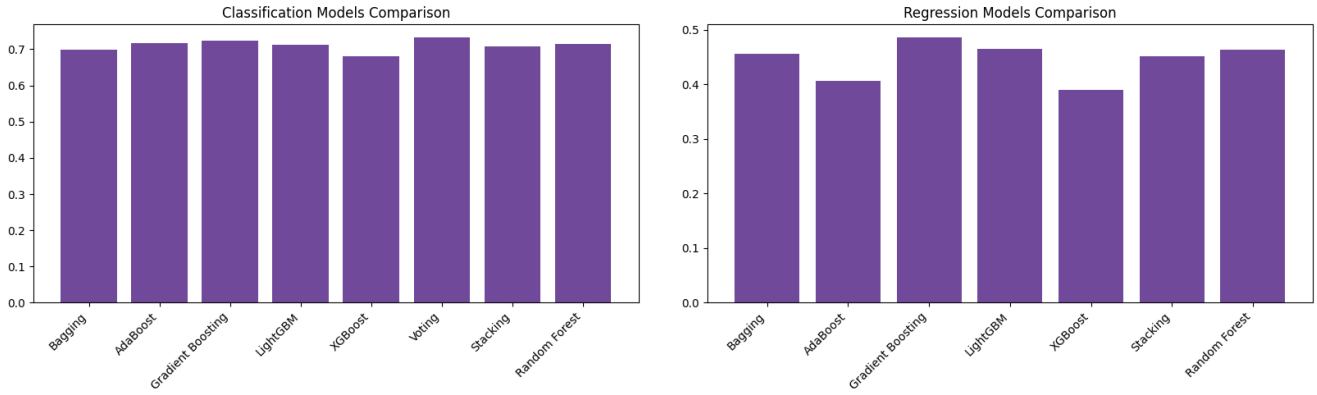



*Figure 1: How each boosting model compares with bagging, and the jump from a single tree to bagging*




*Notebook charts: single tree vs bagging (left) and the four boosting classifiers (right)*




*Notebook charts: classification accuracy (left) and regression R² score (right) of all models*

## 2 Comparison of tree-based boosting algorithms

Each of the four builds trees sequentially. Their test results on both tasks are in the table.

| Model | Accuracy | MAE | RMSE | R² | Note |
|---|---|---|---|---|---|
| AdaBoost | 0.716 | 8.57 | 10.52 | 0.407 | 0.656 accuracy with 1 learner, rising to 0.721 at 25 |
| **Gradient Boosting** | **0.723** | **7.85** | **9.81** | **0.485** | Best on both tasks |
| LightGBM | 0.712 | 8.03 | 10.00 | 0.465 | Runner-up regressor; trained in 0.69 s |
| XGBoost | 0.681 | 8.54 | 10.68 | 0.390 | Last on both tasks (default settings) |

| | |
|---|---|
| **Best model** | Gradient Boosting posted the smallest errors (MAE 7.85, RMSE 9.81) and the top scores on both tasks. |
| **Tuning check** | For Gradient Boosting, a learning rate of 0.1 with 100 estimators gave the best accuracy (0.723). Learning rates of 0.01 and 0.2 scored 0.711 and 0.712, while 50 and 150 estimators scored 0.717 and 0.719. |
| **XGBoost** | With default settings it ranked last on both tasks (accuracy 0.681, R² 0.390). |

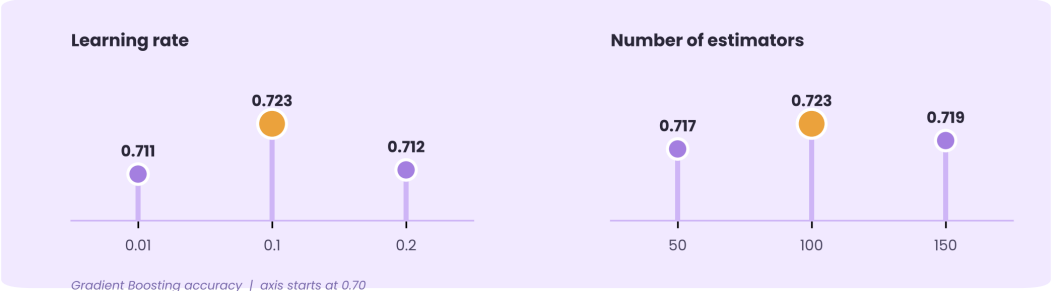



*Figure 2: Tuning check for Gradient Boosting, with the best setting highlighted*

## 3 Advantages of voting and stacking

| | |
|---|---|
| **Voting** | Simple to use, needs no additional model to train, and blends different model types so their errors differ. Hard voting reached 0.732 accuracy, the best classification result in the notebook; soft voting trailed at 0.679. |
| **Stacking** | A meta-learner works out how much weight each base model deserves, and the same idea serves classification and regression. It scored 0.709 accuracy and $R^2$ 0.452. |
| **Trade-off** | Both rely on several models, so they are heavier and harder to explain than a single model. Stacking did not surpass Gradient Boosting on either task here. |

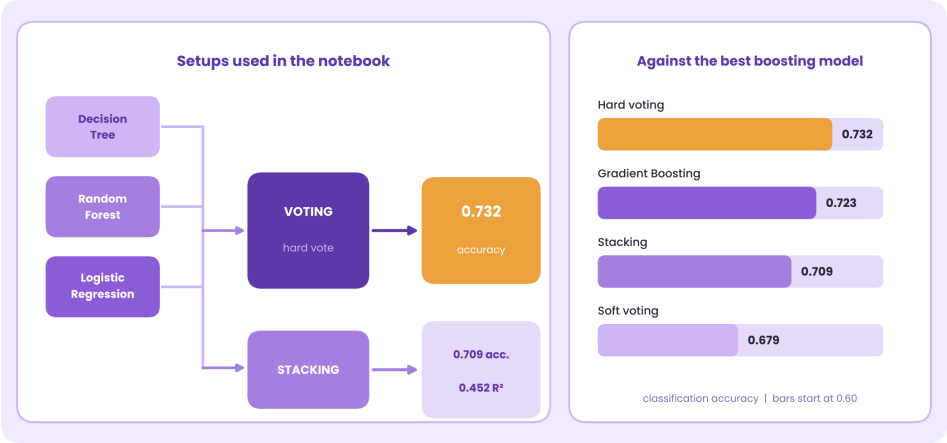



*Figure 3: The voting and stacking setups, with their test results beside Gradient Boosting*

## 4 Final model recommendation for deployment

**CLASSIFICATION | DEPLOY**

### Voting Classifier (hard)

Decision Tree + Random Forest + Logistic Regression

**Accuracy 0.732**

**REGRESSION | DEPLOY**

### Gradient Boosting Regressor

100 estimators, learning rate 0.1 (default)

**MAE 7.85 | RMSE 9.81 | $R^2$ 0.485**

| | |
|---|---|
| **Why these models** | Each posted the best test score for its task in the final comparison (Q30). |
| **Close call** | Voting led Gradient Boosting by just 9 students out of 1,040, and hard voting produces no probability scores. Where probabilities are needed, the Gradient Boosting Classifier (0.723) is a near-equal alternative. |
| **Limits** | Results come from a single train-test split, and an $R^2$ of 0.485 leaves about half of the score variation unexplained. Predictions should assist teachers, not replace their judgement. |

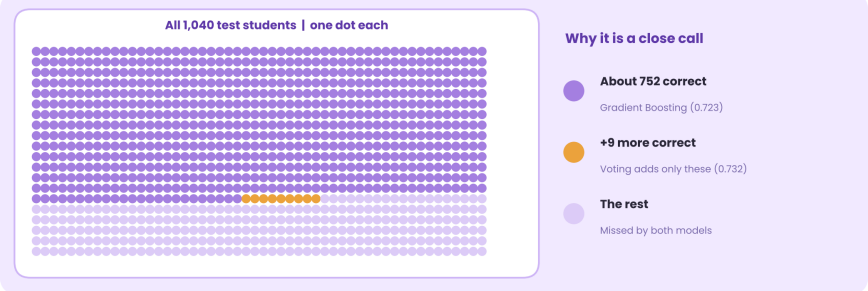



*Figure 4: Voting versus Gradient Boosting across all 1,040 test students (approximate counts from the reported accuracies)*